Bayesian Graphical Modeling

Context

What is psychometrics? 2025

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bayesiangraphicalmodeling.com

Session setup

Part I: History

image: Flaticon.com

Session setup

Part I: History

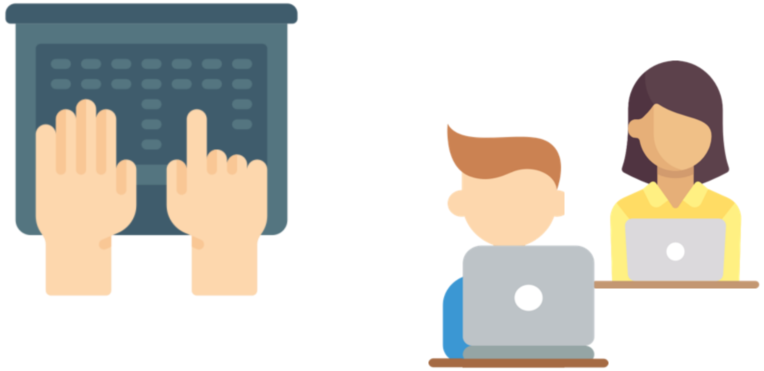
Part II: Theory

image: Flaticon.com

Session setup

Part I: History

Part II: Theory



Part III: Tutorial / Practical

image: Flaticon.com

Session setup

Part I: History

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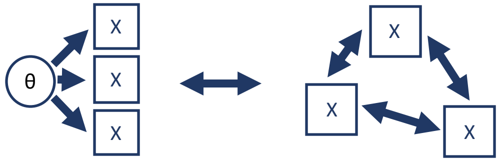
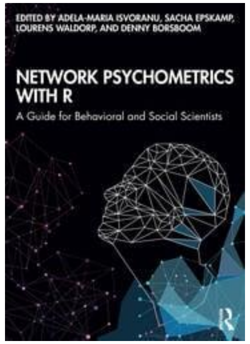
History (What is **network** psychometrics?)



Network **modeling** in psychology

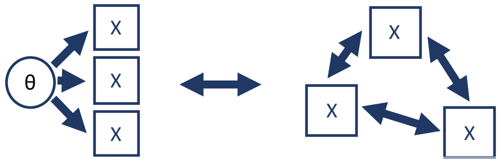
$$P(\mathbf{X} = \mathbf{x}) = \frac{\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i-1} \mu_{ic}\,\mathbb{I}(x_i = c) + \sum_{i>j}\sigma_{ij}x_i x_j\right)}{\sum_{\mathbf{x}}\exp\left(\sum_{i=1}^{p}\sum_{c=1}^{M_i-1} \mu_{ic}\,\mathbb{I}(x_i = c) + \sum_{i>j}\sigma_{ij}x_i x_j\right)}$$

Network **models** in psychology



Network **analysis** in psychology

History (What is **network** psychometrics?)



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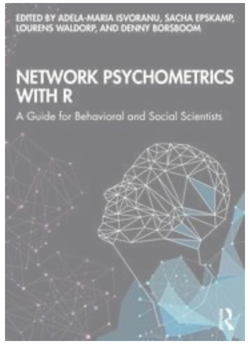
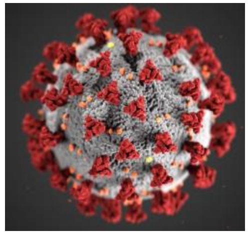
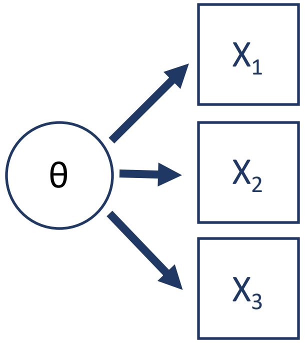


Network **analysis** in psychology

Latent variable model vs network model

Disease analogy





Latent variable model

Latent variable model vs network model

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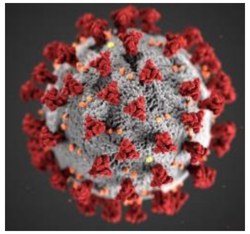
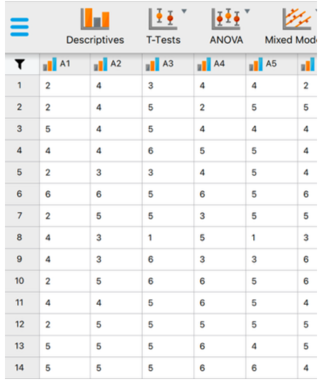


Psychological data



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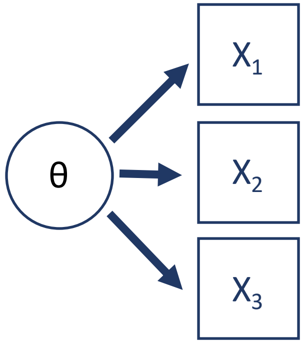
2	2	4	5	2	5	5
3	5	4	5	4	4	4
4	4	4	6	5	5	4
5	2	3	3	4	5	4
6	6	6	5	6	5	6
7	2	5	5	3	5	5
8	4	3	1	5	1	3
9	4	3	6	3	3	6
10	2	5	6	6	5	6
11	4	4	5	6	5	4
12	2	5	5	5	5	5
13	5	5	5	6	4	5
14	5	5	5	6	6	4

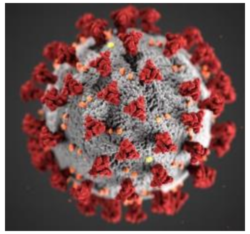


Latent variable model

No conclusive evidence for single cause

Latent variable model vs network model

Disease analogy



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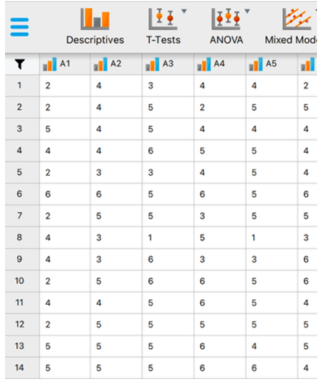

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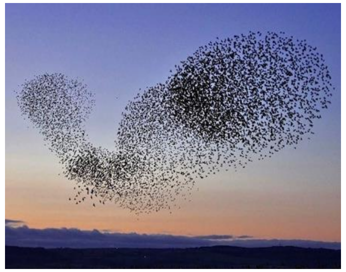
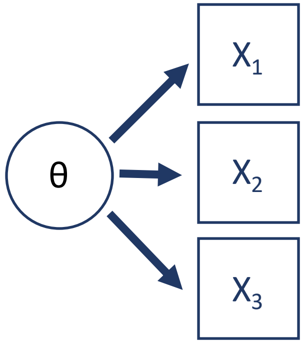
Flocking analogy





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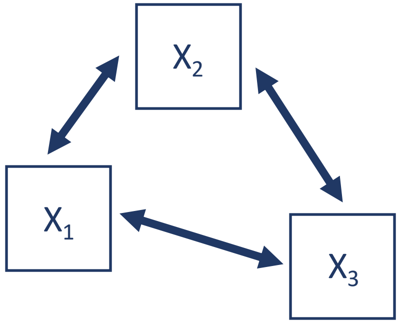

Network model

Literature development

Robinaugh et al. (2018): **732** papers published between 2008 and 2018 on networks.
→ **363** papers on psychiatric phenomena including a network approach

Huth et al. (in press): **4,756** papers published between 2010 and 2023 on networks.
→ **1,423** papers on empirical network analysis of psychometric variables

Further reading

Huth, K. B. S., Haslbeck, J. M. B., Keetelaar, S., van Holst, R. J., & Marsman, M. (in press). Statistical Evidence in Psychological Networks: A Bayesian Analysis of 294 Networks from 126 Studies. *Nature Human Behavior.*

Marsman, M., & Rhemtulla, M. (2022). Guest Editors' introduction to the special issue "Network Psychometrics in Action": Methodological innovations inspired by empirical problems. *Psychometrika*, *87*, 1-11.

Robinaugh, D. J., Hoekstra, R. H. A., Toner, E. R., & Borsboom, D. (2020). The network approach to psychopathology: A review of the literature 2008–2018 and an agenda for future research. *Psychological Medicine*, 50 , 353–366.

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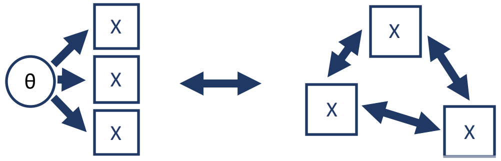

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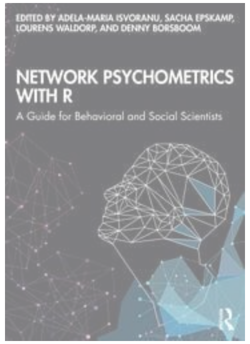

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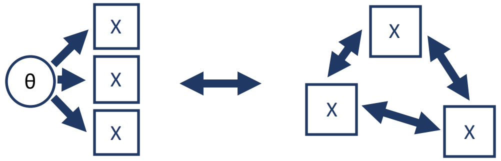

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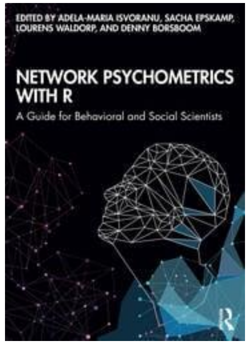

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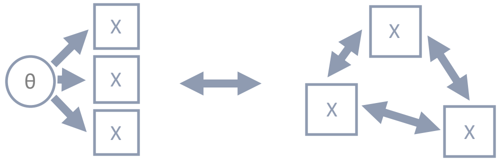

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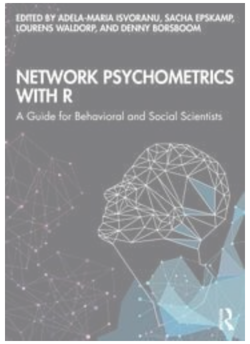
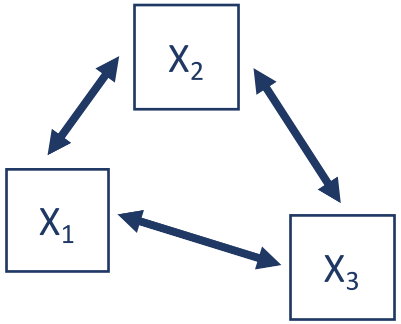


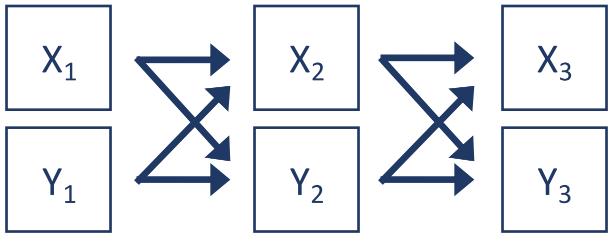
Network **analysis** in psychology

Two distinct research designs (and models)

Cross-sectional



Longitudinal



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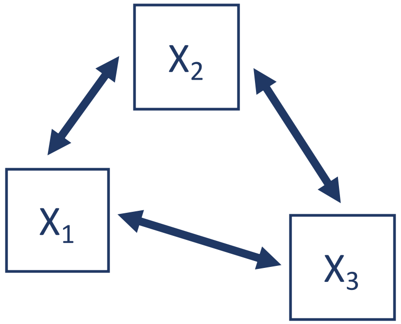
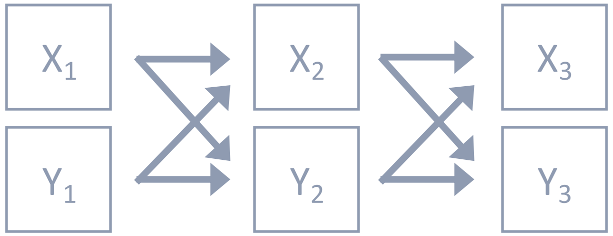


Type of Variable	Model
Binary	Ising model
Continuous	Gaussian graphical model
Mixed	Mixed graphical model

Longitudinal



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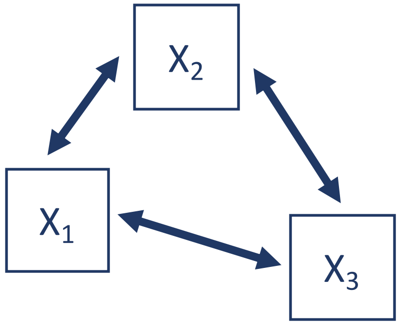

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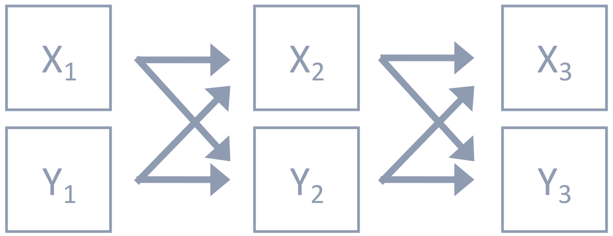

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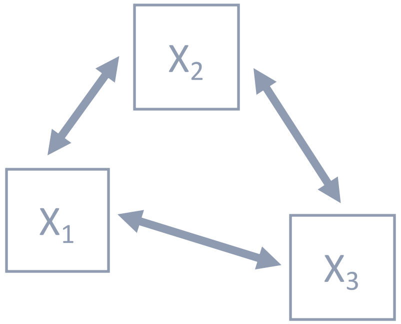

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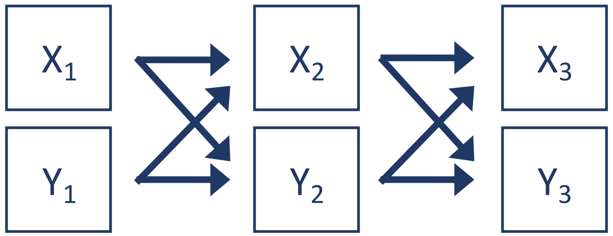

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mixed	mgm (n=1)

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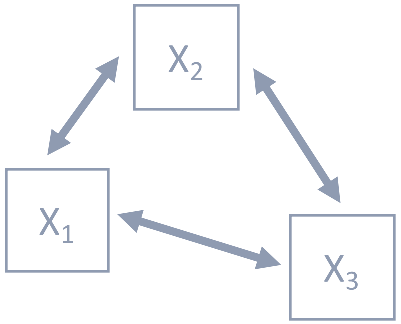

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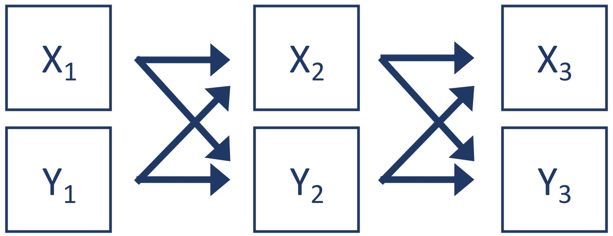

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RELATED	TV-Var, (ml)HMM, Gimme, continuous time models

Ordinal Markov random Field

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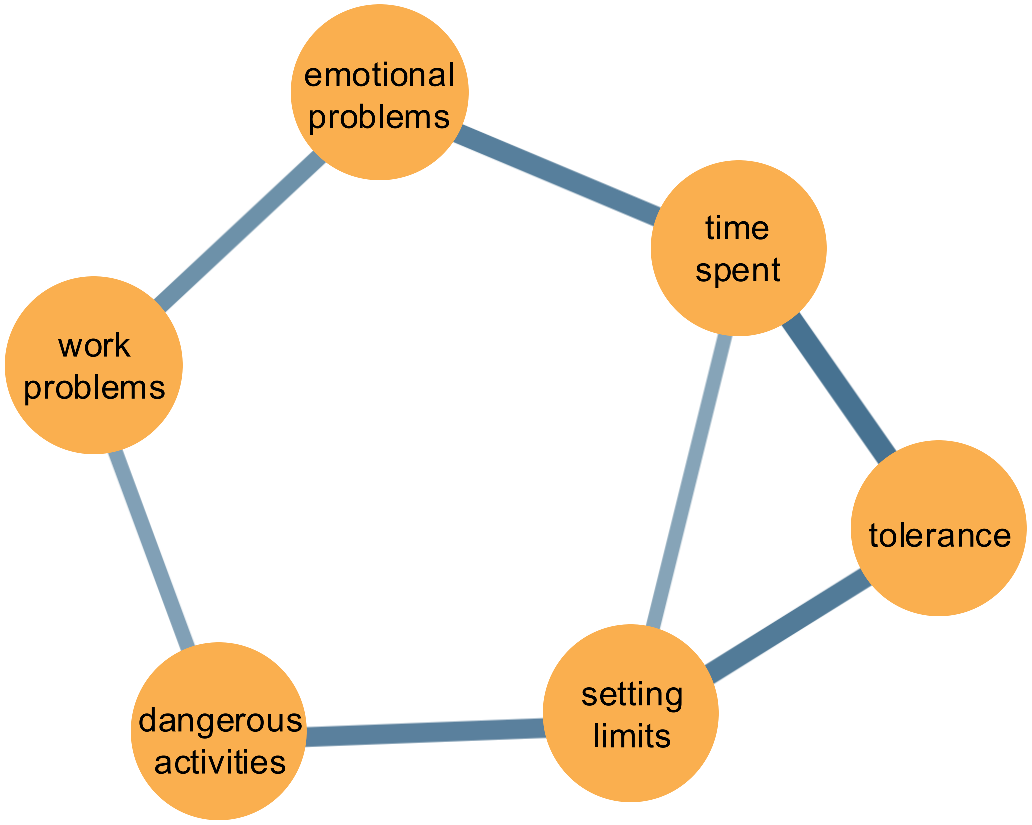

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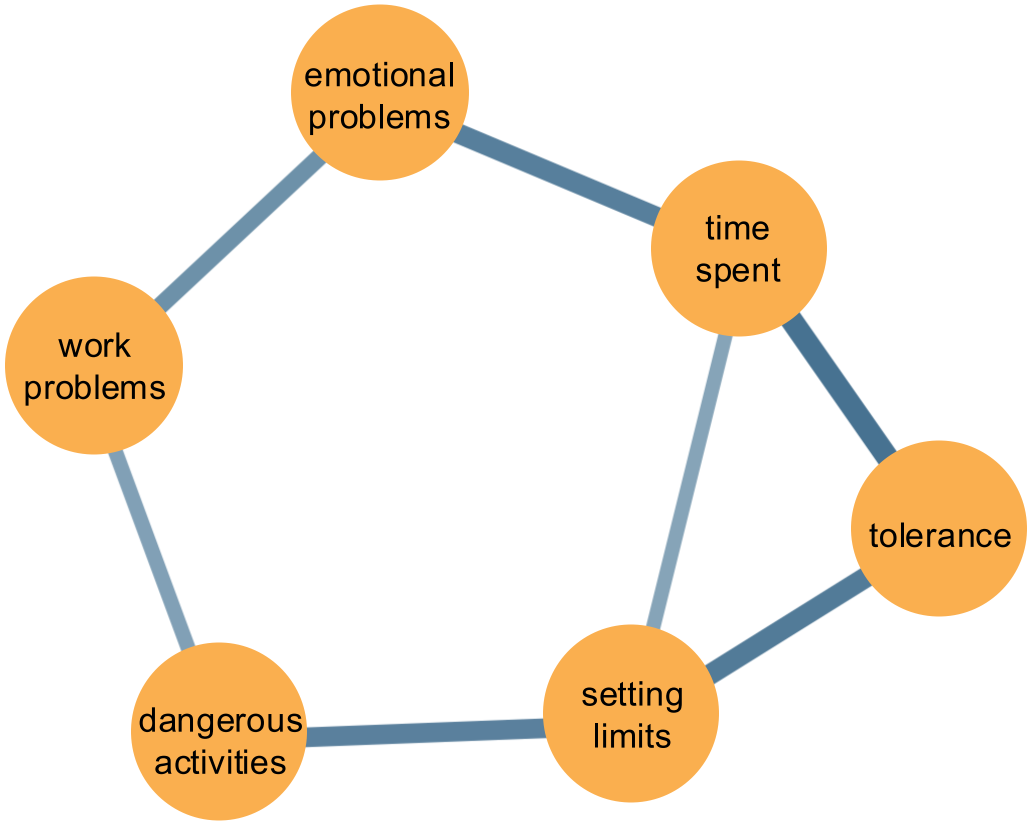

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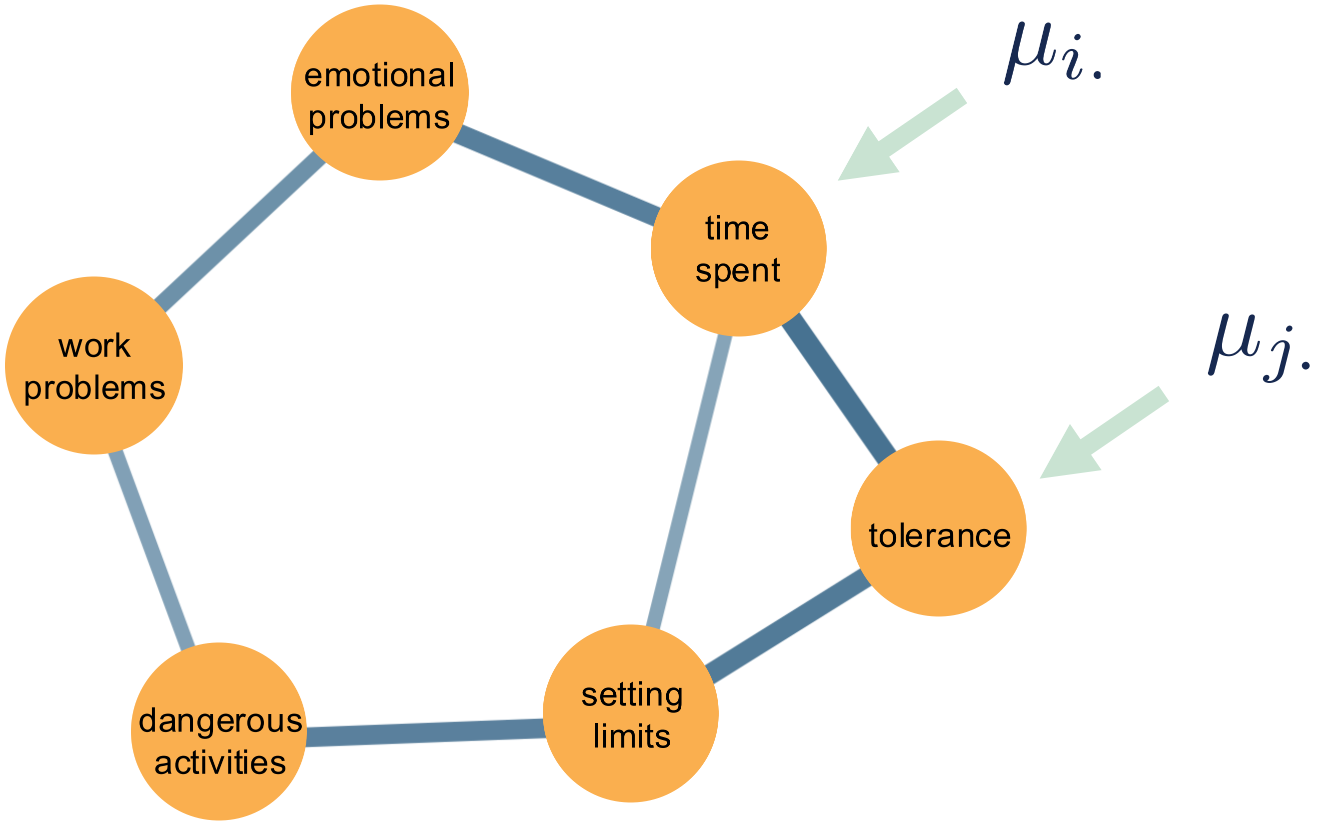

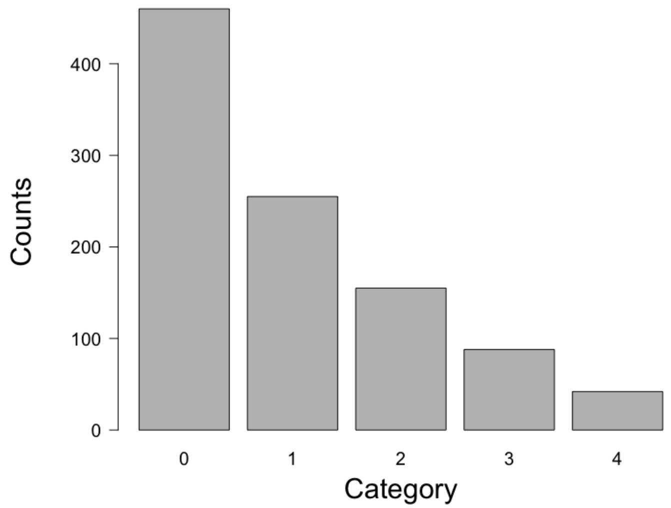

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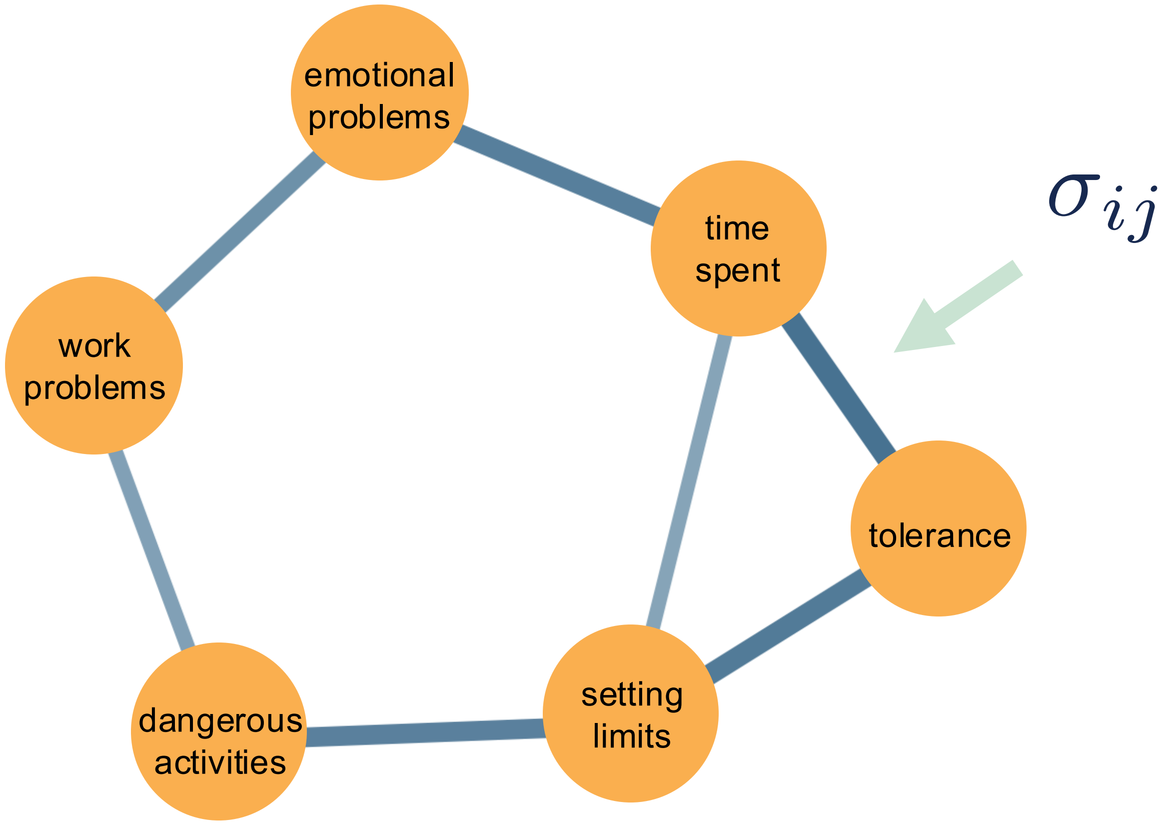

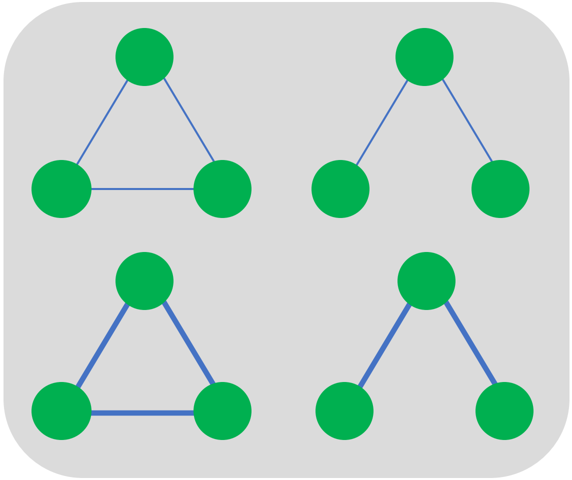

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Further reading

Epskamp, S., Haslbeck, J., Isvoranu, A.-M., & van Borkulo, C. (2022). Pairwise Markov random fields. In A.-M. Isvoranu, S. Epskamp, L. Waldorp, & D. Borsboom (Eds.), *Network Psychometrics with R*. Routledge.

Haslbeck, J., Ryan, O., van der Maas, H., & Waldorp, L. (2022). Modeling change in networks. In A.-M. Isvoranu, S. Epskamp, L. Waldorp, & D. Borsboom (Eds.), *Network Psychometrics with R*. Routledge.

Marsman, M., van den Bergh, D., & Haslbeck, J. (2025). Bayesian analysis of the ordinal Markov random field. *Psychometrika, 90, 146-182.*

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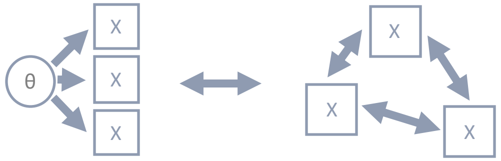

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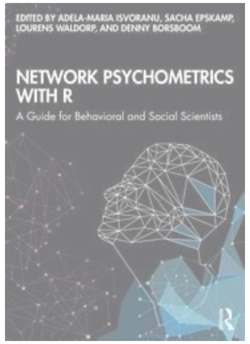

Network **analysis** in psychology

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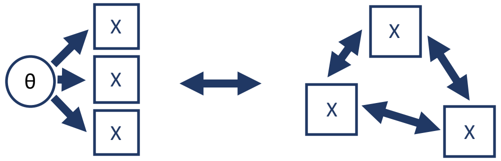

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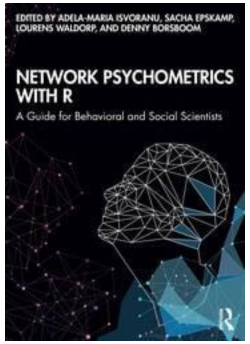

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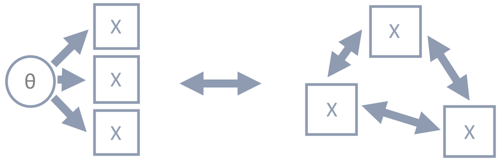

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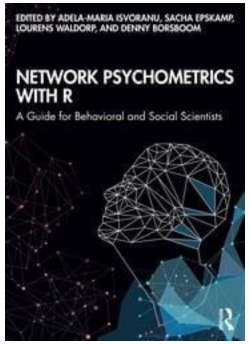
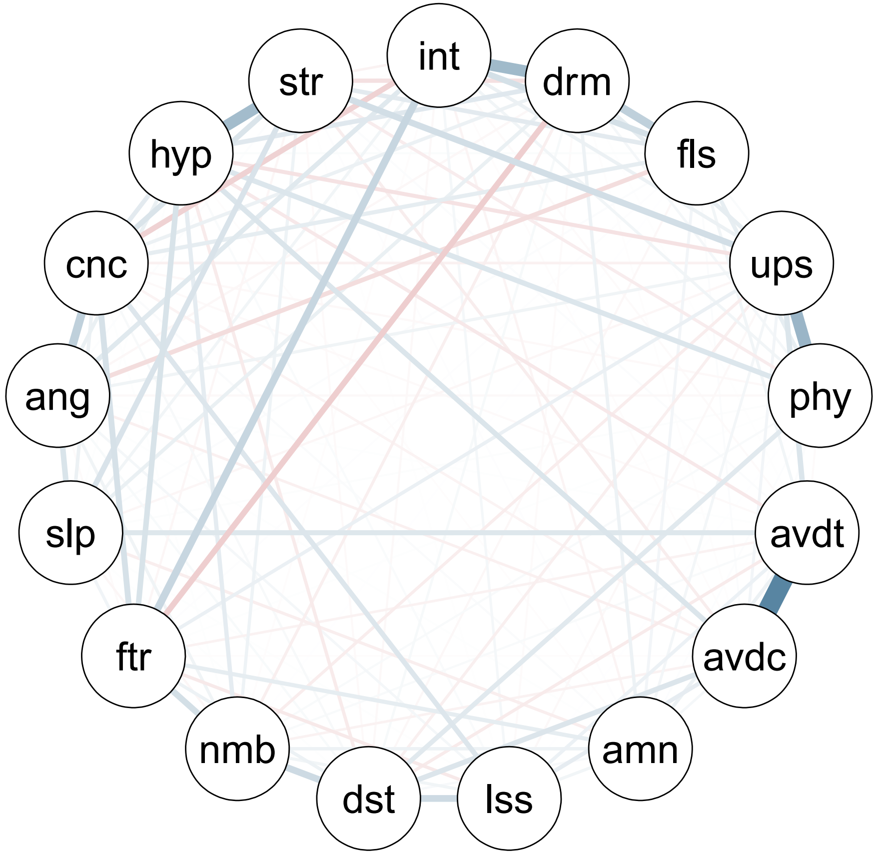


Network **analysis** in psychology

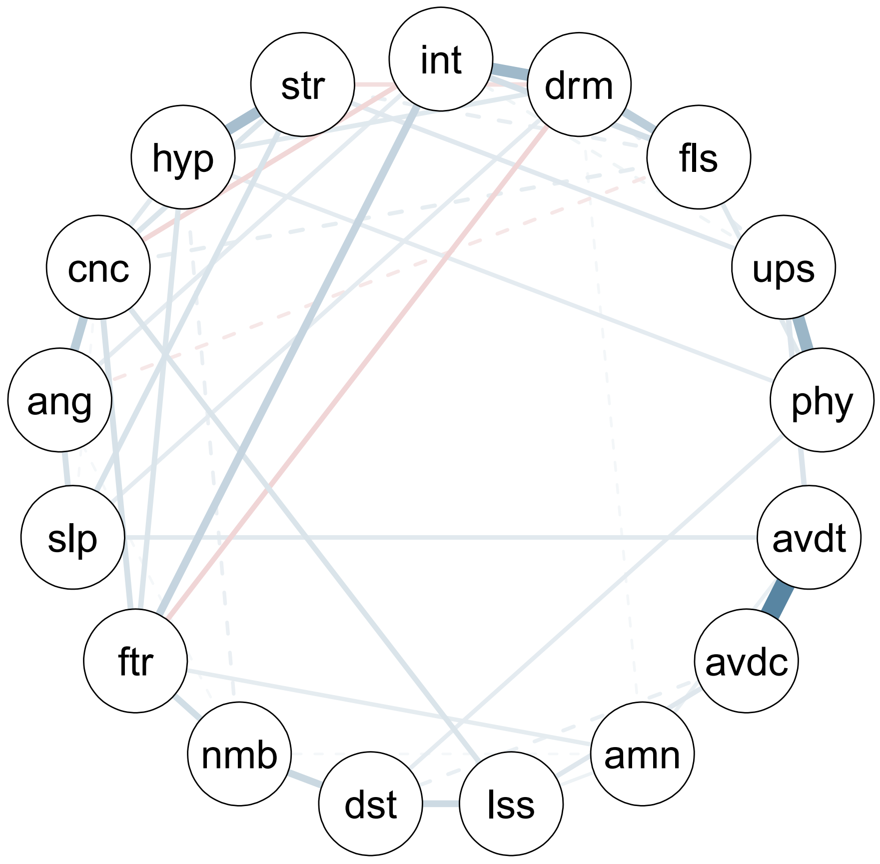
Network estimation



Available methods
Frequentist
Semi-Bayesian (i.e., regularization)
Bayesian

```
library(easybgm)
library(bgms)
fit = bgm(x = Wenchuan, edge_selection = FALSE)
plot_network(fit, layout_avg = "circle", legend = FALSE)
```

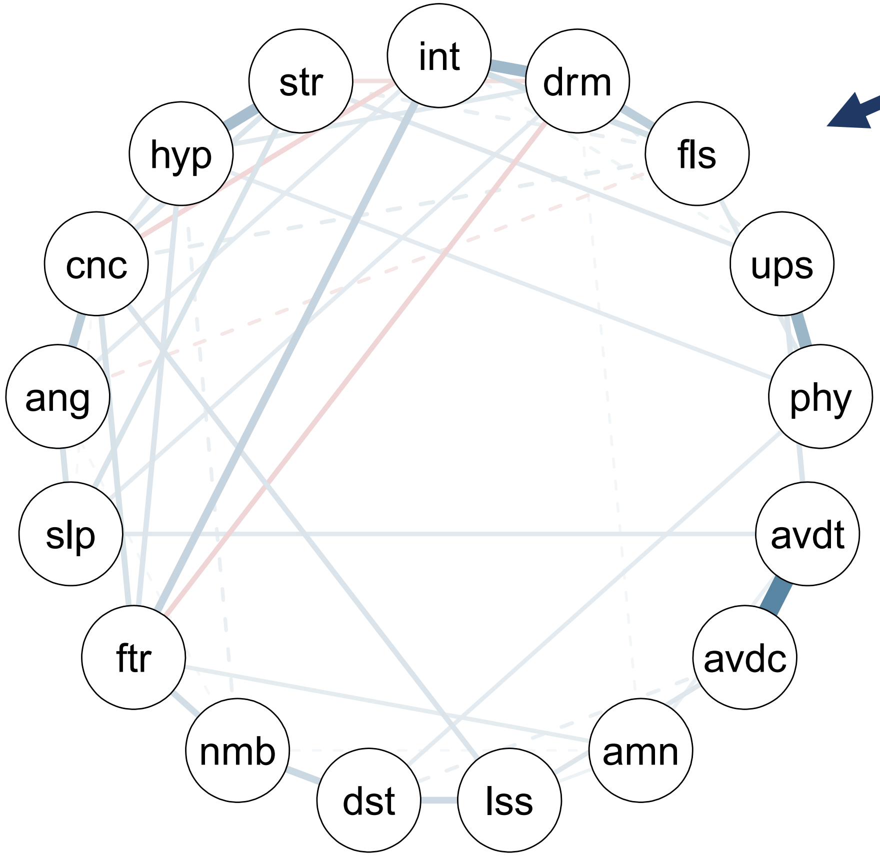
Model Selection



Available methods
Frequentist (i.e., AIC, BIC, etc.)
Semi-Bayesian (empirical Bayes)
Bayesian (Bayes factor)

```r
library(easybgm)
library(bgms)
fit = bgm(x = Wenchuan, edge_selection = TRUE)
plot_network(fit, layout_avg = "circle", legend = FALSE)
```

Model Selection

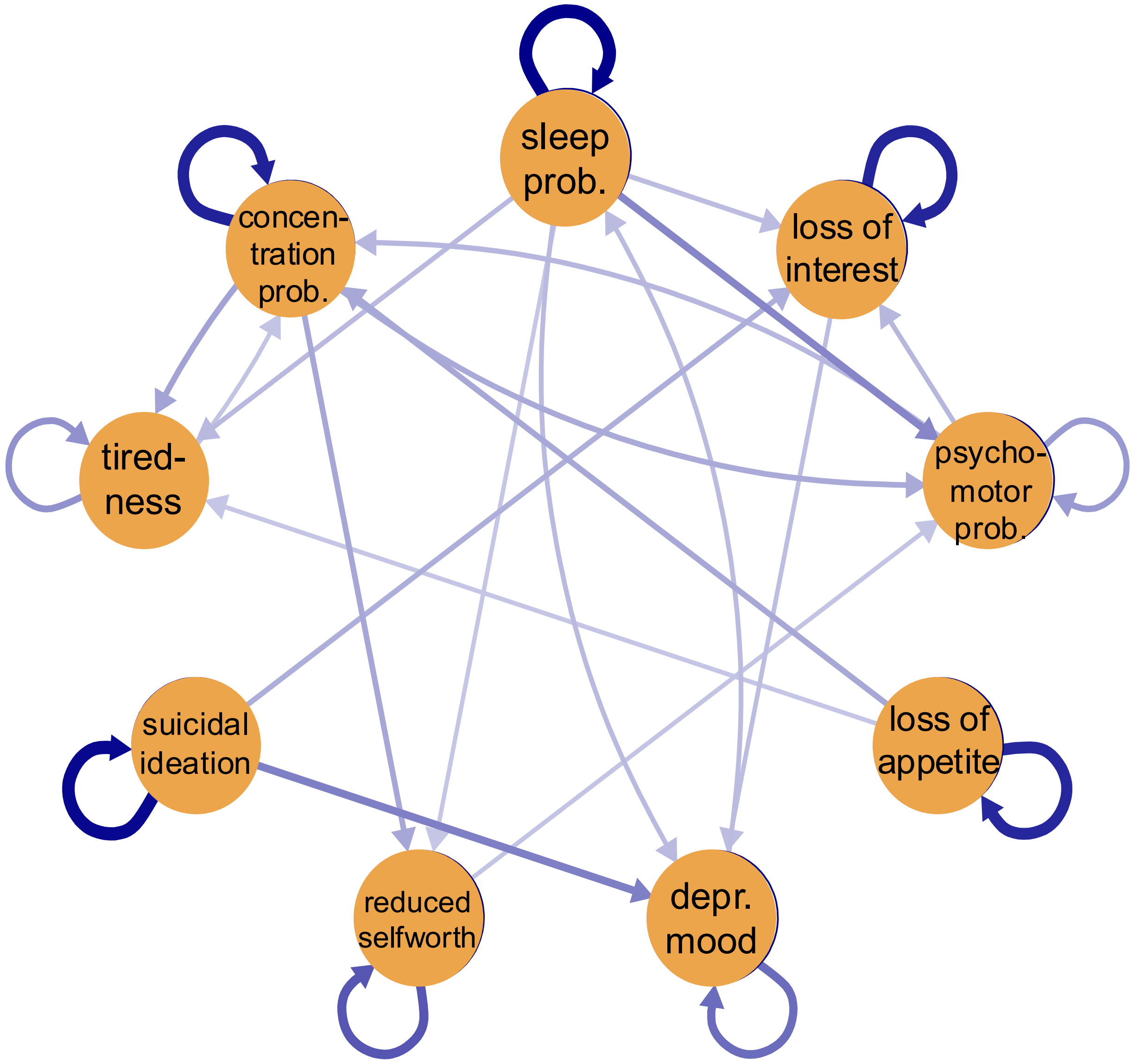


Testing
Cross-Sectional: Conditional independence
Longitudinal: Granger causality

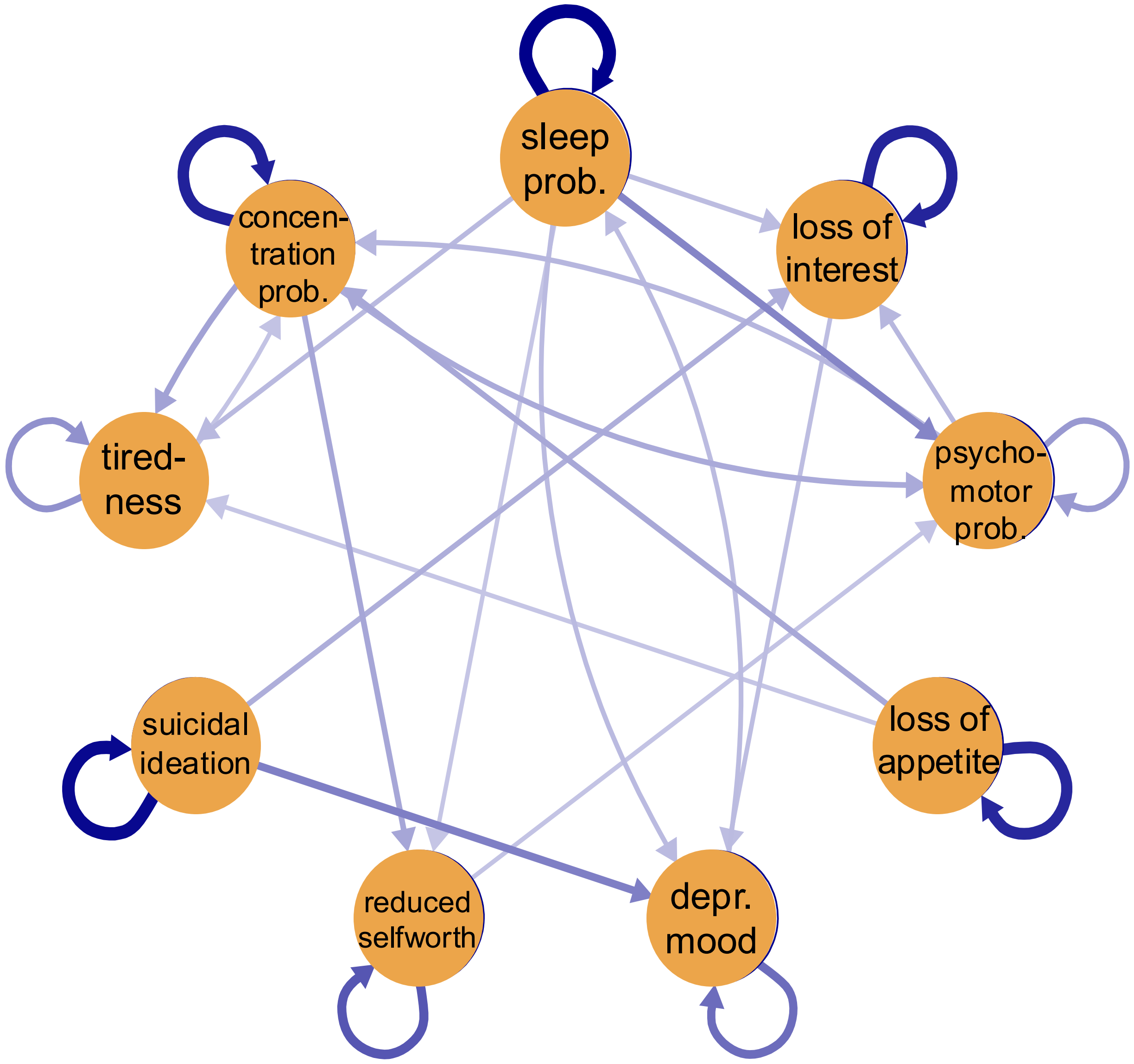
Available methods
Frequentist (i.e., AIC, BIC, etc.)
Semi-Bayesian (empirical Bayes)
Bayesian (Bayes factor)

```
library(easybgm)
library(bgms)
fit = bgm(x = Wenchuan, edge_selection = TRUE)
plot_network(fit, layout_avg = "circle", legend = FALSE)
```

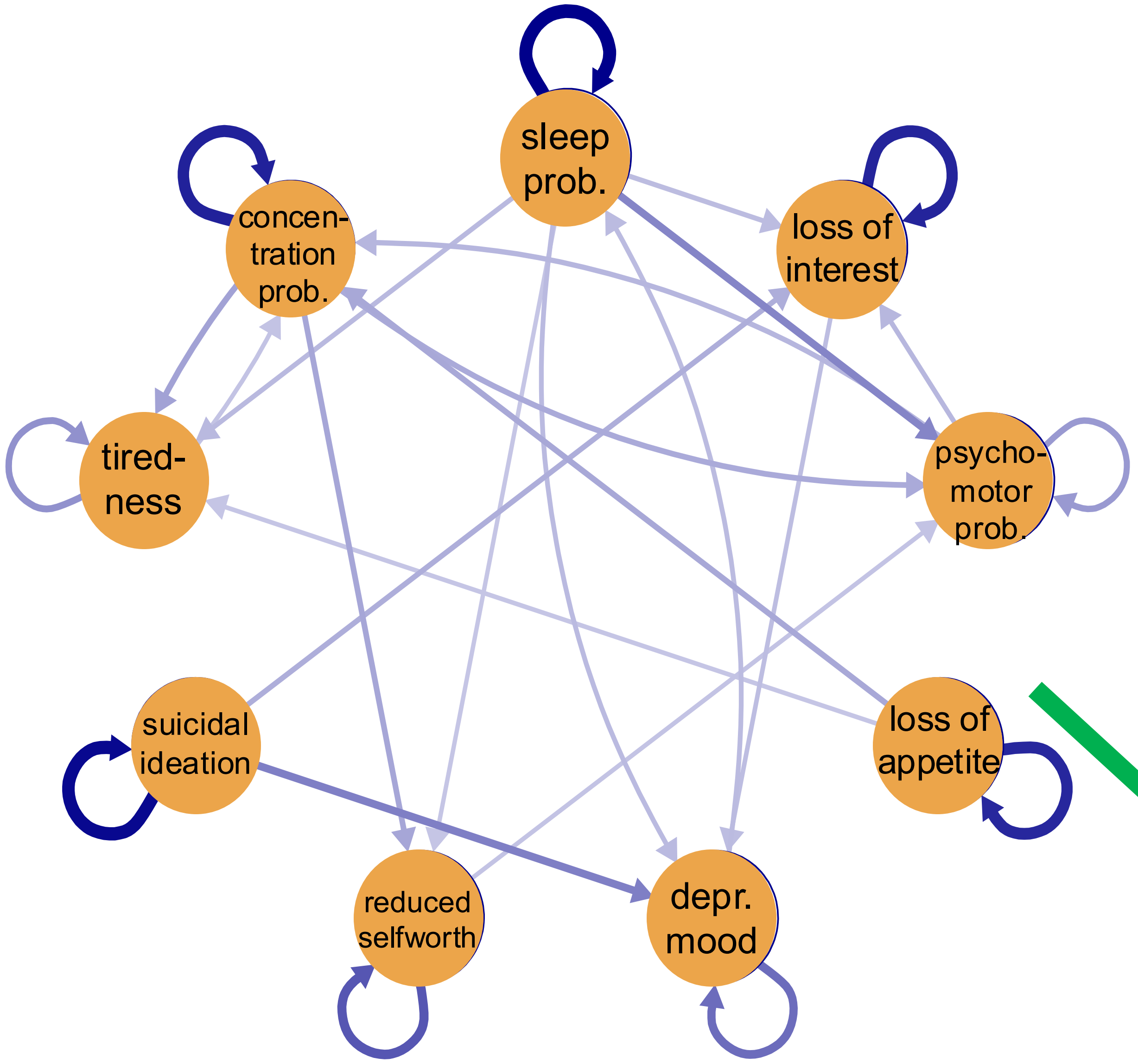
Edge configuration uncertainty is huge!

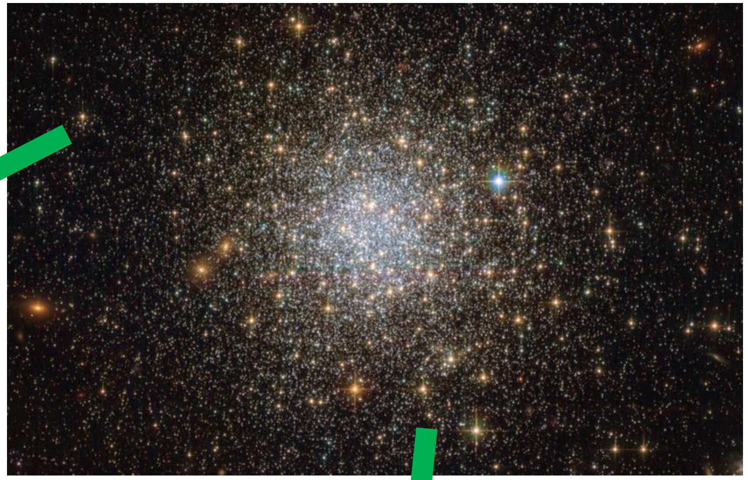


Edge configuration uncertainty is huge!



Edge configuration uncertainty is huge!



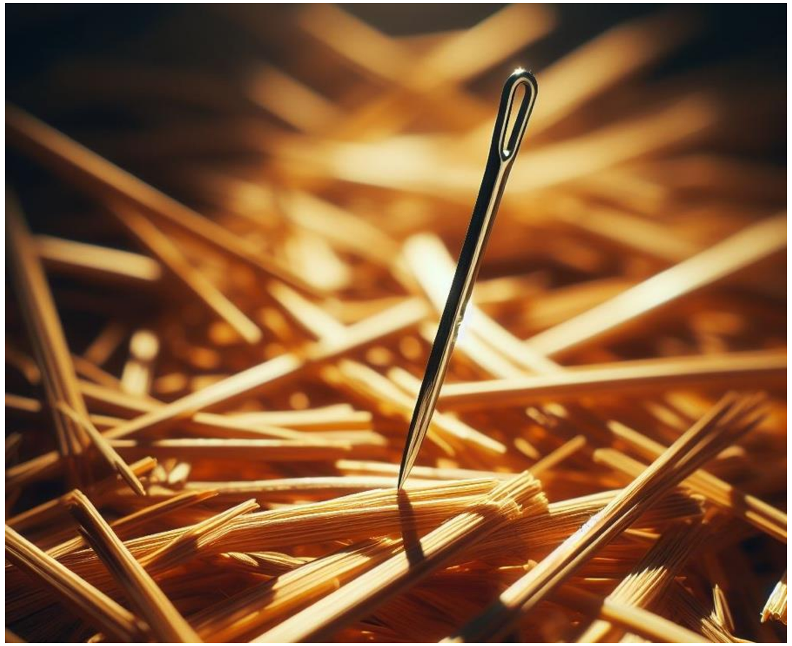


$\approx 1\text{e}{+}24$

$2^{(9\times 9)} \approx 2.42\text{e}{+}24$

Edge configuration uncertainty is huge!

Personal conviction:
Network selection in network psychometrics is useless.



Further reading

Blanken, T., Isvoranu, A.-M., & Epskamp, S. (2022). Estimating network structures using model selection. In A.-M. Isvoranu, S. Epskamp, L. Waldorp, & D. Borsboom (Eds.), *Network Psychometrics with R*. Routledge.

Oravecz, Z., & Vandekerckhove, J. (in press). Quantifying Evidence for—and against—Granger Causality with Bayes Factors. *Multivariate Behavioral Research*.

Sekulovski, N., Keetelaar, S., Huth, K., Wagenmakers, E.-J., van Bork, R., van den Bergh, D., & Marsman, M. (2024). Testing conditional independence in psychometric networks: An analysis of three Bayesian methods. *Multivariate Behavioral Research, 59, 913-933*.

Williams, D., Rhemtulla, M., Wysocki, A., & Rast, P. (2019). On nonregularized estimation of psychological networks. *Multivariate Behavioral Research*, *54*, 719–750.

Session setup

Part I: History

Part II: Theory

image: Flaticon.com

Intermezzo

Open R and run the following code:

```
library(ltm)
#?WIRS
data <- as.matrix(WIRS)

library(IsingFit)
m1 <- IsingFit(data, family='binomial', plot=FALSE)

library(qgraph)
qgraph(m1$weiadj,fade = FALSE)
```

1. Describe what you see;
2. Why is there an edge missing between item 2 and item 3?